SmartRoute AI (Route Wise)

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TARGET: Production MVP

Product & Market Requirements Document (PRD & MRD) — Developer-First Intelligent Traffic-Aware Routing Engine

Executive Summary: Modern routing services default to raw travel time or naive shortest-distance algorithms, ignoring business-specific tolerances for traffic congestion, driver stress, and unpredictable delays. **SmartRoute AI (Route Wise)** delivers an API-first routing intelligence platform and real-time visualization dashboard that evaluates up to 3 candidate driving paths, scores them using a configurable multi-criteria algorithm, categorizes traffic congestion (Green/Yellow/Red), and provides transparent mathematical reasoning for every recommendation.

1. Market Opportunity & Problem Statement

The Market Gap: Fleet operators, on-demand couriers, and mobility developers lose millions annually to urban congestion bottlenecks. Incumbent navigation APIs (e.g., Google Maps, Mapbox) offer closed-box routing without exposed weighting formulas or explicit rationale. Developers cannot penalize severe bottlenecks without building complex post-processing pipelines from scratch.

TAM: $14.2B	SAM: $2.8B	SOM: $350M
Global GIS & Routing Logistics APIs	Urban Delivery & Fleet Routing Devs	Hackathons & Independent Logistics SaaS

2. Target Audience & Customer Personas

- **Primary (Fleet & Dispatch Engineers):** Need deterministic, automated route scoring that minimizes driver delay penalties and fuel variance across multi-vehicle operations.
- **Secondary (On-Demand Couriers & Delivery Startups):** Require sub-second API recommendations with transparent rationale to share with couriers and dispatch managers.
- **Tertiary (Independent Developers & Hackathons):** Need seamless plug-and-play REST endpoints, high-fidelity offline fallbacks, and instantaneous API key governance.

3. Competitive Landscape & Value Proposition

Feature / Dimension	Google Maps API	Mapbox Navigation	SmartRoute AI (Route Wise)
Ranking Customization	None (Opaque defaults)	Limited profiles	**Full (Configurable multi-criteria weights)**
Recommendation Rationale	Not provided	Not provided	**Mathematical scoring & human-readable tags**
Congestion Classification	Visual only	Speed levels	**Independent 3-tier severity (Green/Yellow/Red)**
Offline / Fallback Resilience	Network dependent	Paid offline packs	**Built-in offline geocoding dictionary**
Key Governance & Dev Portal	Complex GCP console	Web dashboard only	**SHA-256 hashed keys + Interactive modal UI**

4. Go-To-Market (GTM) & Monetization Model

Distribution Strategy: Developer-first bottom-up adoption via GitHub, Postman public workspaces, and hackathon showcases. **Tiered Model:** (1) *Developer Free*: 1,000 req/month, rate-limited to 60 req/min; (2) *Growth Fleet*: $0.003/query with custom weight profiles; (3) *Enterprise Dedicated*: SLA guarantees, on-prem geocoding, multi-waypoint batch optimizations.

PRODUCT REQUIREMENTS DOCUMENT (PRD) — PART 1

1. Product Vision, Scope & Core Personas

1.1 Product Vision & Strategic Scope

SmartRoute AI provides an enterprise-grade, developer-friendly traffic intelligence API coupled with a reactive React/Leaflet dashboard. The platform's core directive is **transparency in routing decisions**: instead of accepting a black-box path, operators can inspect why a route was selected (e.g., avoided 12 min gridlock for only 2.1 extra km).

Persona	Role & Context	Primary Needs & Frustrations	SmartRoute AI Value
Dan **Backend Dev**	Building dispatch backend for urban courier platform.	Tired of brittle map APIs with high quotas and unexpected route shifts.	Clean REST endpoints, predictable error codes, offline fallback demo mode.
Olivia **Ops Dispatcher**	Oversees 40+ delivery drivers in high-density metropolitan zones.	Drivers stuck in gridlock during peak hours despite map apps reporting 'optimal'.	Configurable delay penalty weight, live auto-refresh dashboard, visual traffic pins.

1.2 System Scope & Component Boundaries

Module	In-Scope (Current Release v1.0)	Out-of-Scope (Future Releases)
Routing Engine	Origin-Destination single-leg routing with up to 3 driving alternatives, real-time traffic delay penalty, and multi-criteria scoring.	Multi-stop traveling salesperson optimization, pedestrian / bicycle modes, toll-booth cost factoring.
Data Resiliency	Transparent offline fallback dictionary with pre-computed polylines and simulated congestion metrics for key metropolitan hubs.	Full global offline vector tiles, real-time crowdsourced incident telemetry ingestion.
Auth & Security	Cryptographic SHA-256 hashed API key authentication with sliding-window rate limiting (100 req/min) and header enforcement.	OAuth2 / OIDC Single Sign-On, role-based multi-tenant organization hierarchies, billing automation.

2. Detailed Functional Requirements (FR)

Req ID	Module	Functional Specification & Acceptance Criteria	Priority
FR-01	**Geocoding Service**	Accepts text queries (e.g., 'San Francisco, CA') or [lat, lon] tuples. Resolves via TomTom Geocoding API with sub-second fallback to local offline coordinate dictionary when unconfigured or offline.	**P0 (Must)**
FR-02	**Multi-Route Ingestion**	Queries TomTom Routing API requesting up to 3 candidate driving paths (maxAlternatives=2). Ingests polyline geometry, travel duration, distance, and real-time traffic delay in seconds.	**P0 (Must)**
FR-03	**Multi-Criteria Ranking**	Computes composite score: Score = (w_time * Duration) + (w_delay * DelayPenalty) + (w_dist * Distance). Default weights: 50% Time, 35% Delay Penalty, 15% Distance. Lowest score ranks #1.	**P0 (Must)**
FR-04	**Severity Engine**	Independent classification of traffic conditions: **Green (Low)**: delay < 3 mins; **Yellow (Moderate)**: 3 to 8 mins delay; **Red (Heavy)**: > 8 mins delay. Color-codes polylines.	**P0 (Must)**
FR-05	**Transparent Rationale**	Generates human-readable badges in API response (e.g., 'Fastest route (24m)', 'Avoids 9m gridlock', 'Slightly longer distance (+1.4 km)'). Explains exact trade-off.	**P1 (Should)**
FR-06	**Key Governance & Auth**	Protects developer endpoints via X-API-Key. Keys hashed with SHA-256 and stored in server/data/keys.json. Includes rate-limiter middleware (100 req/min per key).	**P0 (Must)**
FR-07	**Interactive Dashboard**	React + Vite SPA featuring Leaflet map, origin/dest swap, preset cities (SF, NYC, London), route summary cards, cURL documentation modal, and key management modal.	**P0 (Must)**
FR-08	**Live Auto-Refresh**	Visual 60-second circular countdown timer that automatically invokes POST /api/routes/refresh. Allows pause, manual trigger, and stale-data notifications.	**P1 (Should)**

PRODUCT REQUIREMENTS DOCUMENT (PRD) — PART 2

2. Technical Specifications, NFRs & Roadmap

2.1 System Architecture & Data Pipeline

The system operates as a decoupled client-server architecture. The frontend (Vite/React 18) communicates over JSON REST with an Express.js backend. Requests pass through three layers: (1) **Security & Validation** (coordinate sanitization, SHA-256 key check, sliding-window rate limiter); (2) **Integration Layer** (TomTom Routing Client with offline dictionary failover); and (3) **Ranking & Severity Core**.

2.2 Core API Endpoints & Interfaces

Method & Route	Header & Body Parameters	Response Payload & Status	SLA / Perf
POST /api/routes/recommend	X-API-Key: <key> { origin, destination, weights? }	200 OK: Ranked routes array with GeoJSON polyline, duration, trafficDelay, score, tags.	< 350ms (TomTom) < 35ms (Fallback)
POST /api/routes/refresh	X-API-Key: <key> { routeIds: [...] }	200 OK: Fresh delay metrics and recalculated scores without re-geocoding coordinates.	< 250ms (Live) < 20ms (Fallback)
GET /api/health	Public (No auth required)	200 OK: { status: 'healthy', tomtom: true, uptime: 10452s }	< 15ms
GET /api/developer/status	X-API-Key: <key>	200 OK: Key metadata, quota used, requests remaining, tier limits.	< 25ms

2.3 Non-Functional Requirements & Success Metrics (KPIs)

Non-Functional Requirements (NFRs)	Target Success Metrics & Business KPIs
• **Availability:** 99.9% uptime with instant offline mock fallback. • **Latency:** p95 API response < 450ms across public networks. • **Security:** Cryptographic SHA-256 API key hashing, zero key leakage in logs. • **Resilience:** Graceful degradation if TomTom limits exceeded. • **Test Coverage:** 100% pass on 22 automated unit/integration tests.	• **Developer Adoption:** 10,000+ API calls within first 90 days of launch. • **Delay Mitigation:** 14% average reduction in route delay vs. raw shortest path. • **Zero-Downtime Fallback:** 100% successful response rate in demo/offline mode. • **Dashboard Performance:** < 1.2s initial load, 60 FPS map panning/zooming. • **Developer Satisfaction (CSAT):** > 4.6 / 5.0 on docs & cURL integration.

2.4 Product Release Milestones & Phased Roadmap

Phase / Target	Scope & Key Deliverables	Deliverable Highlights & Gate Criteria
Phase 1: MVP Current (Q4 2026)	Core Recommendation Engine, Leaflet UI & Fallback System	Node/Express API, TomTom integration, offline dictionary, weighted ranking, React 18 dashboard, 22 unit tests.
Phase 2: Expansion Q1 2027	Multi-Stop Dispatch & Dynamic Webhook Alerts	Traveling Salesperson (TSP) multi-waypoint optimization, automated webhook alerts on severe traffic spikes, SDKs for Python & Go.
Phase 3: Scale Q2 2027	EV Fleet Battery Optimization & Weather Overlay	Elevation & weather impact modeling, EV charging station waypoints, enterprise team permissions, billing portal integration.

2.5 Technical Risk Analysis & Mitigation Matrix

Identified Risk / Threat	Severity	Mitigation Strategy & Built-In Safeguard
TomTom API Rate Quota Exhaustion	Medium	Automatic fallback to offline mock dictionary; sliding window rate limiter protects upstream quota.
Sudden Traffic Spike Discrepancy	Low	Client 60-second polling cadence with manual refresh trigger and UI stale-data warning badges.
Developer Key Compromise	High	One-way SHA-256 hash storage in keys.json; secrets redacted from server logs and client bundles.